Create Great Graphics and Visualisation with DSLs

This article introduces readers to some of the domain specific languages (DSLs) for interactive graphics and data visualisation. It explores DSLs like Processing, Vega and Graphviz, going into depth about their capabilities with basic examples.

A domain-specific language (DSL) is a computer language targeted for a specific domain. It improves developer productivity as it is easier to program. It breaks the communication barrier between domain experts and programmers since the language and its syntax are more native to the former. Though DSLs have been around for a long time, they have gained popularity of late. UNIX has hosted a lot of little languages like *awk, sed, pic, grap,* etc, which were developed for a specific application domain. They are called mini languages in UNIX and recent DSLs can be considered glorified mini languages. Unlike UNIX mini languages, modern DSLs are more expressive and even provide APIs for other general-purpose languages. Some of the widely used DSLs are SQL for querying databases, MATLAB for mathematical computing, CSS for styling Web pages, UML for software design, etc.

The success of DSLs in many domains has resulted in their growing popularity in newer domains. Graphics and visualisation is one such field in which many new DSLs are emerging. In this field, the focus is on creating approaches to convey abstract information in a way that is easily understandable to humans. In the early days, graphics and visualisation were limited to statistical charts and static images.